

BAYESIAN ANALYSIS OF AN OUTLIER MODEL IN LIFE TESTING

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Summary : Life estimation based on a few ordered observations from the exponential distribution is considered in this paper for situations where the first failure occurs at a very early stage of the life test and is suspected to be a possible "outlier". A new parametric model for this outlier problem is proposed and the Bayes estimators of the mean life and reliability are obtained for a suitable prior density on the parameter space and the squared error loss function. The $100(1-\alpha)\%$ highest posterior density interval (HPDI) for the estimate of the mean life is obtained. A numerical example is discussed.

1. Introduction

In a recent paper, Bhattacharya & Singh (1979) discussed the problem of life estimation for situations where the first failure occurs at a very early stage of a life test and is suspected to be an "early failure", that is, an "outlier". The outlier model considered therein was as follows:

For a preassigned r , let $x_1, x_2, \dots, x_r$ be the first r ordered observations available with $(n-r)$ survivors at the termination of the

life test as in Epstein & Sobel (1953). We assume that the smallest observation x_1 can possibly be an outlier, that is, it comes from a population with probability density

$$(1.1) \quad \frac{1}{\nu\theta} f\left(\frac{x}{\nu\theta}\right), \quad 0 < \nu \leq 1, \quad \theta > 0$$

while each of the remaining x_i ($i=2,3,\dots,r$) comes from the population with density

$$(1.2) \quad \frac{1}{\theta} f\left(\frac{x}{\theta}\right), \quad \theta > 0$$

where $f(x) = \exp(-x)$, $0 < x < \infty$.

When $\nu = 1$, this is called the "homogeneous case" and for $0 < \nu < 1$, this is referred to as the "non-homogeneous case". Here actually x_1 is an outlier and ν provides a measure of what may be called a "spuriosity coefficient". This model is akin to what Barnett & Lewis (1978) call "labelled slippage" model and is essentially different from the "exchangeable model" of other authors (see Kale & Sinha (1971) and Sinha (1973)), where they assume that the outlying observation is equally likely to be any of the sample observations. This assumption of equal probability is somewhat unrealistic as the discordant observation, if any, is most likely to be the extreme one (see Barnett & Lewis, *op. cit.* p. 37). In our model, one can test whether x_1 is an outlier by testing a preliminary hypothesis:

$$(1.3) \quad H : \nu = 1$$

against the alternative

$$(1.4) \quad H' : 0 < \nu < 1.$$

Bhattacharya & Singh (*op. cit.*) obtained a UMP test of H based on the likelihood ratio and proposed a pretest estimator of θ depending upon the result of the preliminary test. Undoubtedly our interest is in estimating θ rather than testing the discordancy of the possible outlier, namely, the test of the nuisance parameter ν . In this paper we consider the Bayesian analysis of the outlier model proposed above under the assumption of suitable prior densities on the parameter space under the squared error loss function.

In the present approach we are not interested in the test of discordancy since the problem of outlier is taken care of quite naturally in a Bayesian context.

2. The Bayes estimator of the mean life

We first write down the joint density of the r ordered observations $\mathbf{x} = (x_1, \dots, x_r)'$ under the condition that the smallest observation x_1 is suspect, that is, it comes from the density (1.1) whilst the rest of the observations come from (1.2). Thus, we are concerned with $[n, B, \tau]$ plan of Gnedenko, *et al.* (1969), the joint density of $x_1, \dots, x_r$ for our model can be written as

$$(2.1) \quad f(x_1, \dots, x_r | \nu, \theta) = \frac{(n-1)!}{(n-r)!} \frac{1}{\theta^r} e^{-T/\theta} \left\{ (n-1) + \frac{1}{\nu} \right\} e^{-x_1/\nu\theta},$$

$$0 < x_1 < x_2 < \dots < x_r < \infty, \theta > 0, 0 < \nu \leq 1$$

where

$$T = \sum_{i=2}^r x_i + (n-r)x_r.$$

Given the observations $x_1, x_2, \dots, x_r$ and considered as a function of θ and ν , the expression (2.1) provides the sample likelihood from which the Bayesian posterior can be obtained in the usual manner by choosing

appropriate priors on the parameter space of (θ, ν) . We take, for example, an inverted gamma prior for θ (see Bhattacharya (1967)) given by the pdf

$$(2.2) \quad g(\theta) = \frac{\mu^\rho}{\Gamma(\rho)} \frac{e^{-\mu/\theta}}{\theta^{\rho+1}}, \quad 0 < \theta < \infty, \quad \mu, \rho > 0.$$

Further we assume that θ and ν are independent *a priori* and that the nuisance parameter ν has a beta density

$$(2.3) \quad g(\nu) = m(1-\nu)^{m-1}, \quad 0 < \nu \leq 1, \quad m = 1, 2, \dots$$

It may be noted that the larger the value of m , the higher the prior density if ν is near zero and faster the density decreases when ν moves away from zero. In such a situation, the probability that the observation with mean $\nu\theta$ is the smallest observation will be large. This is compatible with our model assumptions.

With the priors in (2.2) and (2.3), the joint prior density of (θ, ν) is

$$(2.4) \quad g(\theta, \nu) = \frac{m\mu^\rho}{\Gamma(\rho)} \frac{(1-\nu)^{m-1} e^{-\mu/\theta}}{\theta^{\rho+1}}, \quad 0 < \theta < \infty, \quad 0 < \nu \leq 1$$

where $\mu, \rho > 0$. It may be noted that g is being used as a generic notation for the prior densities.

Now the sample evidence at (2.1) is combined with the prior information at (2.4) to yield the joint posterior density

$$(2.5) \quad g^{\forall}(\theta, \nu | x) = \frac{h(\nu, \theta | x)}{\int_0^\infty \int_0^1 I(\theta | x) d\theta d\nu}, \quad 0 < \theta < \infty, \quad 0 < \nu \leq 1$$

where

$$(2.6) \quad h(\nu, \theta | x) = \begin{cases} \frac{1}{\theta^{r+\rho+1}} e^{-(T+\mu)/\theta} (1-\nu)^{m-1} \left\{ (n-1) + \frac{1}{\nu} \right\} e^{-x_1/\nu\theta} & 0 < \theta < \infty, \quad 0 < \nu \leq 1, \quad \mu, \rho > 0 \\ 0, & \text{elsewhere} \end{cases}$$

and

$$(2.7) \quad I(\theta | x) = \int_0^1 h(\nu, \theta | x) d\nu, \quad 0 < \theta < \infty.$$

Hence, the marginal posterior density of θ is given by

$$(2.8) \quad g^\forall(\theta | x) = I(\theta | x) / \int_0^\infty I(\theta | x) d\theta, \quad 0 < \theta < \infty.$$

Note that all inferences about θ can be drawn from this posterior density. Under the assumption of a squared error loss function, the Bayes estimator of θ is simply the posterior mean of (2.8) and is given by

$$(2.9) \quad \hat{\theta} = \int_0^\infty \theta I(\theta | x) d\theta / \int_0^\infty I(\theta | x) d\theta.$$

To compute first $I(\theta | x)$ consider

$$(2.10) \quad \theta^{r+\rho+1} e^{(T+\mu)/\theta} I(\theta | x) = \int_0^1 (1-\nu)^{m-1} \left\{ (n-1) + \frac{1}{\nu} \right\} e^{-x_1/\nu\theta} d\nu$$

$$= \sum_{k=0}^{m-1} (-1)^k \binom{m-1}{k} \left\{ (n-1) \int_1^\infty \frac{e^{-x_1 u/\theta}}{u^{k+2}} du + \int_1^\infty \frac{e^{-x_1 u/\theta}}{u^{k+1}} du \right\}$$

$$= \sum_{k=0}^{m-1} (-1)^k \binom{m-1}{k} \left\{ (n-1) E_{k+2} \left(\frac{x_1}{\theta} \right) + E_{k+1} \left(\frac{x_1}{\theta} \right) \right\}$$

where

$$(2.11) \quad E_n(z) = \int_1^\infty \frac{e^{-zu}}{u^n} du, \quad n = 0, 1, 2, \dots; \operatorname{Re} z > 0.$$

In particular, for $n=1$

$$(2.12) \quad E_1(z) = \int_Z^\infty \frac{e^{-u}}{u} du \quad \operatorname{Re} z > 0$$

is the exponential integral frequently used in astrophysics and nuclear physics (cf. Erdélyi, (1953)).

From (2.10) we obtain

$$(2.13) \quad I(\theta|x) = \frac{e^{-(T+\mu)/\theta}}{\theta^{r+\rho+1}} \sum_{k=0}^{m-1} (-1)^k \binom{m-1}{k} \left\{ (n-1) E_{k+2}\left(\frac{x_1}{\theta}\right) + E_{k+1}\left(\frac{x_1}{\theta}\right) \right\}.$$

In order to evaluate the integrals introduced in (2.9), we use the relationship between $E_n(z)$ and the incomplete gamma function $\Gamma(d, z)$, and the Laplace transform of Erdélyi (*op. cit.*, p. 138, formula (8)) to obtain the following result:

$$(2.14) \quad \int_0^\infty e^{-st} t^{\beta-n} E_n(t) dt = \frac{\Gamma(\beta-n+1)}{\beta(1+s)^{\beta-n+1}} {}_2F_1[1, \beta-n+1; \beta+1; \frac{s}{1+s}]$$

provided that $\operatorname{Re} \beta > 0$, $\operatorname{Re}(\beta-n+1) > 0$ and $\operatorname{Re} s > -\frac{1}{2}$, where ${}_2F_1[\cdot]$ is the Gauss hypergeometric function. The conditions mentioned here hold for the following integrals and one obtains

$$(2.15) \quad \int_0^\infty I(\theta|x) d\theta = \frac{\Gamma(r+\rho)}{(T+\mu+x_1)^{r+\rho}} \sum_{k=0}^{m-1} (-1)^k \binom{m-1}{k} \left\{ \frac{(n-1)}{(r+\rho+k+1)} {}_2F_1[1, r+\rho; r+\rho+k+2; y] + \frac{1}{(r+\rho+k)} {}_2F_1[1, r+\rho; r+\rho+k+1; y] \right\},$$

where

$$y = (T+\mu)/(T+\mu+x_1).$$

Similarly we obtain

$$(2.16) \quad \int_0^\infty \theta I(\theta|x) d\theta = \frac{\Gamma(r+\rho-1)}{(T+\mu+x_1)^{r+\rho-1}} \sum_{k=0}^{m-1} (-1)^k \binom{m-1}{k} \left\{ \frac{(n-1)}{(r+\rho+k)} \right. \\ \left. {}_2F_1[1, r+\rho-1; r+\rho+k+1; y] + \frac{1}{(r+\rho+k-1)} {}_2F_1[1, r+\rho-1; r+\rho+k; y] \right\}$$

and

$$(2.17) \quad \int_0^\infty \theta^2 I(\theta|x) d\theta = \frac{\Gamma(r+\rho-2)}{(T+\mu+x_1)^{r+\rho-2}} \sum_{k=0}^{m-1} (-1)^k \binom{m-1}{k} \left\{ \frac{(n-1)}{(r+\rho+k-1)} \right. \\ \left. {}_2F_1[1, r+\rho-2; r+\rho+k; y] + \frac{1}{(r+\rho+k-2)} {}_2F_1[1, r+\rho-2; r+\rho+k-1; y] \right\}$$

provided that $r + \rho > 2$. Hence the Bayes estimator $\hat{\theta}$ for θ is obtained from (2.9) by substituting from (2.15) and (2.16). The variance of the posterior distribution $g^V(\theta|x)$ is then obtained by using the formula

$$(2.18) \quad V = \left[\int_0^\infty \theta^2 I(\theta|x) d\theta / \int_0^\infty I(\theta|x) d\theta \right] - (\hat{\theta})^2$$

in conjunction with (2.9), (2.15) and (2.17).

3. Reliability estimation

The reliability function for the exponential density $\frac{1}{\theta} f(\frac{x}{\theta})$ is $R = R_\theta(t) = e^{-t/\theta}$, ($t > 0$). The estimation of this function is of considerable interest in the literature on life testing (see Bhattacharya, 1967). Under the assumption of the squared error loss function, the Bayes estimator of reliability is simply the posterior expectation of $e^{-t/\theta}$ with respect to $g^V(\theta|x)$. Hence the Bayes estimator of reliability is given by

$$(3.1) \quad \hat{R} = \int_0^\infty e^{-t/\theta} I(\theta|x) d\theta / \int_0^\infty I(\theta|x) d\theta.$$

The integral in the numerator is evaluated by using (2.13) and (2.14). Thus we obtain

$$(3.2) \quad \int_0^\infty e^{-t/\theta} I(\theta|x) d\theta = \frac{\Gamma(r+\rho)}{(T+\mu+x_1+t)^{r+\rho}} \sum_{k=0}^{m-1} (-1)^k \binom{m-1}{k} \cdot \\ \left\{ \frac{(n-1)}{(r+\rho+k+1)} {}_2F_1[1, r+\rho; r+\rho+k+2; v] \right. \\ \left. + \frac{1}{(r+\rho+k)} {}_2F_1[1, r+\rho; r+\rho+k+1; v] \right\},$$

where $v = (T + \mu + t)/(T + \mu + x_1 + t)$.

Substitution of (2.15) and (3.2) in (3.1) leads to the Bayes estimator of reliability.

3.1 Numerical example

A random sample of size 10 simulated from the exponential distribution (1.2) with $\theta = 2$ is given below:

$$z_1 = 0,624, \quad z_2 = 0,03, \quad z_3 = 0,106, \quad z_4 = 0,729, \quad z_5 = 1,542, \\ z_6 = 1,046, \quad z_7 = 2,815, \quad z_8 = 1,126, \quad z_9 = 5,921 \text{ and } z_{10} = 3,630.$$

Notice that the smallest observation in the sample is 0,03. Suppose that we replace 0,03 by 0,003 which is fairly small and can be treated as an outlier. Then the first $r = 7$ observations in ascending order are:

$$x_1 = 0,003, \quad x_2 = 0,106, \quad x_3 = 0,624, \quad x_4 = 0,729, \\ x_5 = 1,046, \quad x_6 = 1,126 \text{ and } x_7 = 1,542.$$

Using this ordered sample and $\rho = 2$, $\mu = 2$ and $m = 2$, the Bayes estimate of θ obtained from (2.9) is

$$(3.1.1) \quad \hat{\theta} = 1,420$$

and the maximum likelihood estimate (see Mann, *et al.*(1974), p. 164) is

$$(3.1.2) \quad \tilde{\theta} = \left\{ \sum_{i=1}^n x_i + (n-r)x_r \right\} / r = 1,40.$$

4. The highest posterior density interval

Let $p(\theta|x)$ denote the posterior density function of θ , given x . An interval I^+ in the parameter space Θ of θ is called an HPD interval of content $(1-\alpha)$ if

$$(a) \quad Pr\{\theta \in I^+ | x\} = 1 - \alpha$$

$$(b) \quad \text{For every } \theta_1 \in I^+ \text{ and } \theta_2 \notin I^+ \\ p(\theta_1|x) \geq p(\theta_2|x).$$

Though we expect the posterior density (2.8) to be unimodal, we discuss the method of finding the HPD interval for the following forms of a posterior density:

- (i) J -shaped such that $p(\theta_1|x) < p(\theta_2|x)$ for $\theta_1 < \theta_2$;
- (ii) U -shaped asymmetric;
- (iii) unimodal asymmetric.

Let ℓ and U denote the lower and upper limits of the $100(1-\alpha)\%$ HPD interval for θ . Then

- (i) If the posterior density function is J -shaped such that $p(\theta_1|x) < p(\theta_2|x)$ for $\theta_1 < \theta_2$, the $100(1-\alpha)\%$ HPDI is one-sided and hence can be determined from

$$(4.1) \quad \Pr\{\theta > \ell \mid x\} = \int_{\ell}^{\infty} p(\theta|x) d\theta = 1-\alpha.$$

- (ii) When the posterior density is U -shaped and asymmetric, one method (see Singh (1985)) for determining the limits is to consider the following two identities:

$$(4.2) \quad p(\ell|x) = p(U|x)$$

$$(4.3) \quad \int_0^{\ell} p(\theta|x) d\theta + \int_U^{\infty} p(\theta|x) d\theta = 1 - \alpha.$$

Those values of ℓ and U that satisfy both (4.2) and (4.3), for given α , will be the lower and upper limits of the $100(1-\alpha)\%$ HPD interval.

- (iii) If the posterior density is unimodal and asymmetric, the HPDI limits will be those which satisfy (4.2) and (4.3) with $(1-\alpha)$ replaced by α in (4.3).

4.1 Numerical example

Given $n = 10$, $r = 7$, $x_1 = 0.003$, $\rho = 2$, $\mu = 2$, $m = 2$ and

$$T = \sum_{i=2}^7 x_i + (n-r)x_r = 9.7990.$$

Then

$$y = (T+\mu)/(T+\mu+x_1) = 0.99975.$$

Let ℓ and U be the lower and upper 95% HPD limits of the estimate $\hat{\theta}$ of θ . The marginal posterior density of θ is given by (2.8). Then the 95% HPD limits will be given by the equations:

$$(4.4) \quad g^V(\ell|x) = g^V(U|x)$$

$$(4.5) \quad \int_0^{\ell} g^V(\theta|x) d\theta + \int_U^{\infty} g^V(\theta|x) d\theta = 0,05.$$

Note from (2.8) that

$$(4.6) \quad g^V(\theta|x) = I(\theta|x) / \int_0^{\infty} I(\theta|x) d\theta.$$

For the above given values, equation (2.15) gives

$$(4.7) \quad \int_0^{\infty} I(\theta|x) d\theta = \frac{\Gamma(9) \times 9,467}{(11,8)^9}$$

and from (2.13) we have

$$(4.8) \quad I(\theta|x) = \frac{e^{-11,8/\theta}}{\theta^{10}} \left\{ 8 E_2 \left(\frac{x_1}{\theta} \right) + E_1 \left(\frac{x_1}{\theta} \right) - 9 E_3 \left(\frac{x_1}{\theta} \right) \right\}.$$

Substituting (4.6), (4.7) and (4.8) in (4.5) it can be seen that the incomplete integrals involved are fairly complex. To overcome this difficulty the exponential integrals $E_i \left(\frac{x_1}{\theta} \right)$, $i = 1, 2, 3$ are approximated for very small values of x_1/θ using the tables given in the "Handbook of mathematical functions" by Abramowitz and Stegun. In this example x_1/θ is 0,0015. Then the approximations are:

$$E_1(0,0015) = 5,9236, \quad E_2(0,0015) = 0,9903, \quad E_3(0,0015) = 0,5.$$

Using these approximations and substituting (4.8), (4.7) and (4.6) in the first term of (4.5), we have

$$(4.9) \quad \int_0^{\ell} g^V(\theta|x) d\theta = \frac{9,3456}{9,4670} \frac{1}{\Gamma(9)} \int_{11,8/\ell}^{\infty} e^{-\nu} \nu^8 d\nu.$$

Similarly, it can be seen that

$$(4.10) \quad \int_U^{\infty} g^V(\theta|x) d\theta = \frac{9,3456}{9,4670\Gamma(9)} \int_0^{11,8/U} e^{-\nu} \nu^8 d\nu.$$

It may be noted that (4.9) and (4.10) involve incomplete gamma functions.

Thus, using (4.4) and the Tables of Incomplete Gamma Function we arrive at

$$(4.11) \quad \frac{0,9872}{\Gamma(9)} \int_{11,8/\ell}^{\infty} e^{-\nu} \nu^8 d\nu = 0,018$$

and

$$(4.12) \quad \frac{0,9872}{\Gamma(9)} \int_0^{11,8/U} e^{-\nu} \nu^8 d\nu = 0,032$$

which yield

$$\ell = 0,72 \quad \text{and} \quad U = 2,71.$$

Thus the 95% HPD interval for θ is $(0,72, 2,71)$.

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